Ethanol Blending in Gasoline – South Korea

July 2026

U.S. GRAINS & BIOPRODUCTS COUNCIL

20 F St. NW, Suite 900 \ Washington, DC 20001
202-789-0789 \ www.grains.org

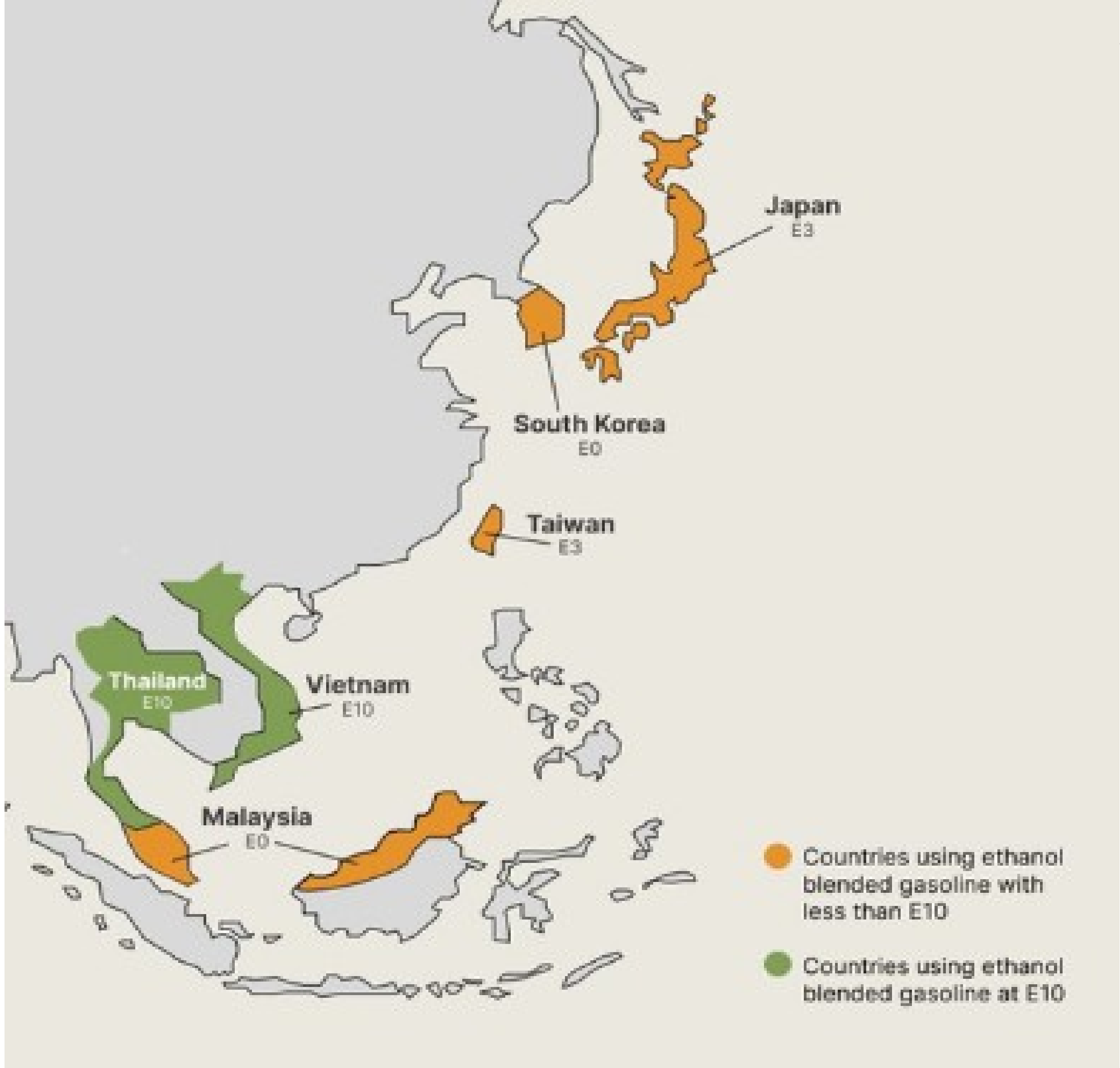
FARO90

www.faro90.com
+52 55 6122 2255

Ethanol Blending in Asia

There are important fuel quality and environmental impact of vehicle emission challenges in these regions.

- The use of ethanol improves gasoline quality and creates flexibility in gasoline production.
- Ethanol use is a cost-effective way to increase gasoline octane and to replace more expensive gasoline components.
- Ethanol contributes to transport decarbonization and air quality improvement.
- There are opportunities across Asia to increase the ethanol blend level and implement new policies on the use of gasoline-ethanol blends.



Seven countries with potential and additional use of ethanol were studied: 1) Gasoline market profiles; 2) Optimization of gasoline blends with ethanol and, 3) Environmental impact of gasolines blended with ethanol.

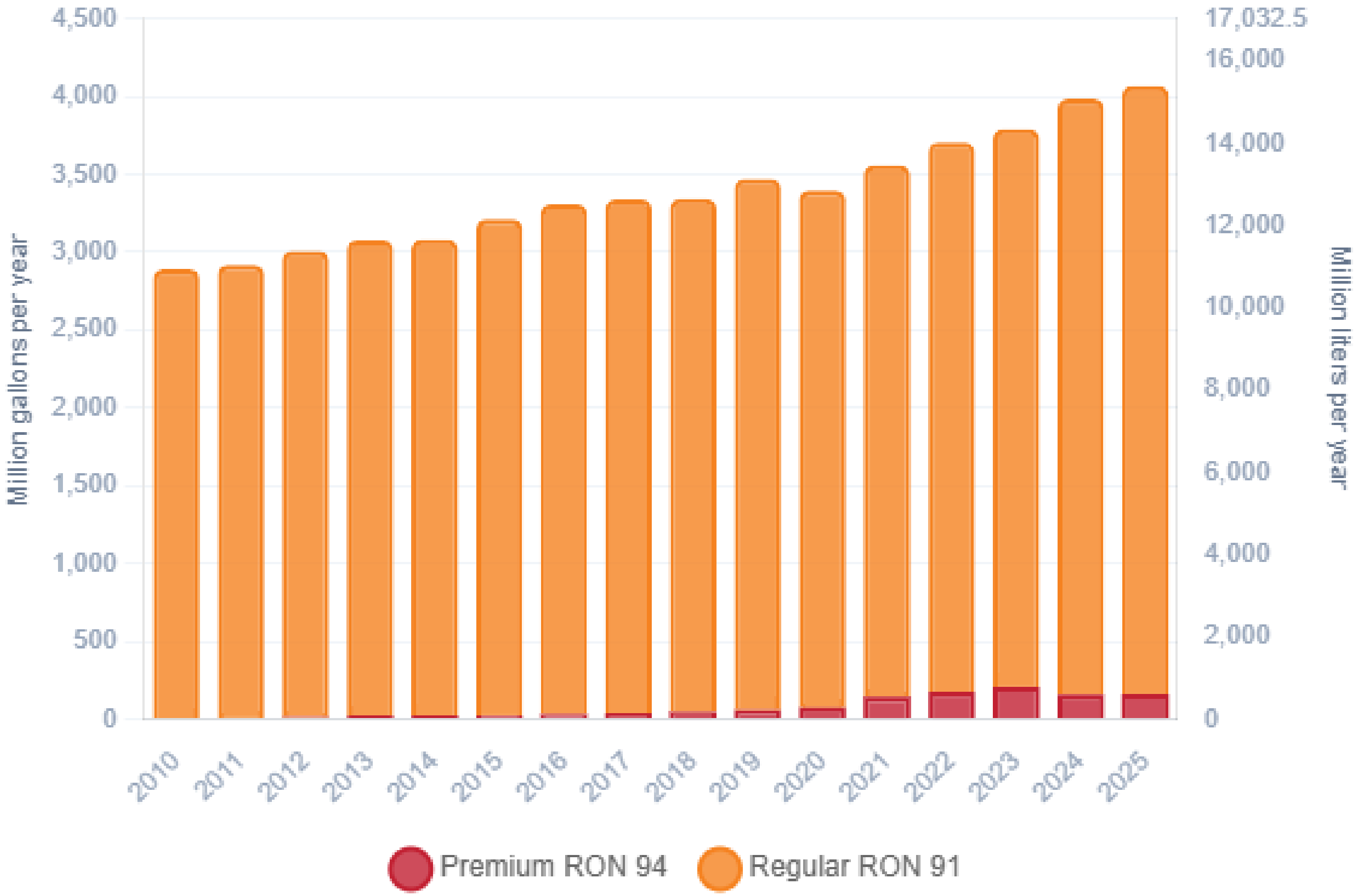
Ethanol Gasoline Blending in South Korea

In 2025, gasoline consumption in South Korea was 15,425 million liters (4,071 million gallons) and growth rate was 2.3%. Gasoline production and net imports were 32,383 and -17,715 million liters (8,555 and -4,680 million gallons) correspondingly. South Korea offers two main grades: RON 91 and RON 94) with an average octane of 91.1 RON.

South Korea has implemented requirements for renewable fuels that are currently met with biodiesel blending. Altough, there is an oxygen requirement, ethanol is not currently blended in gasoline. Efforts to implement pilot programs have been stalled because of a lack of consensus and reluctance of different actors in the petroleum industry.

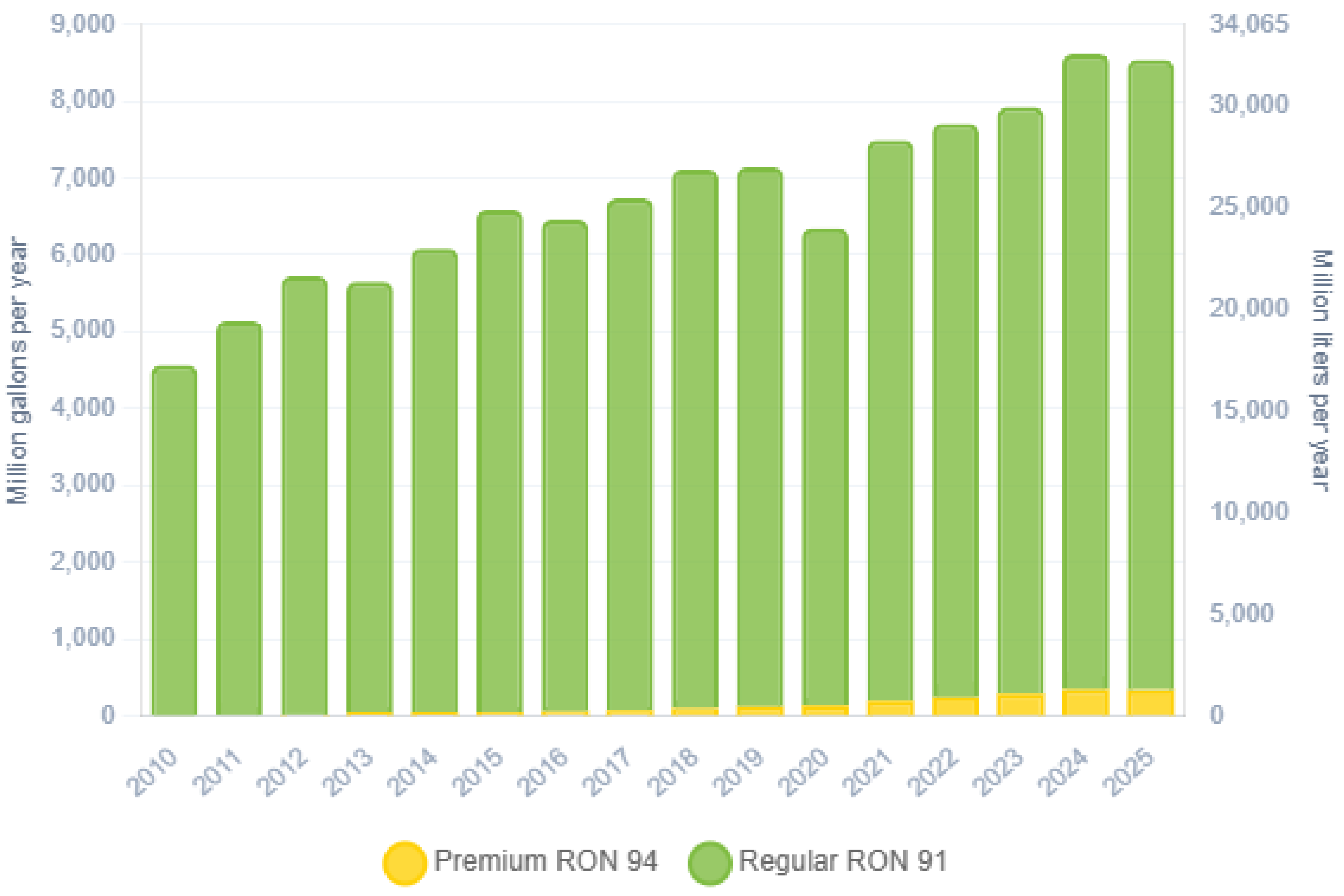
Source: Korean Energy Economics Institute, IEA

Gasoline Demand in South Korea



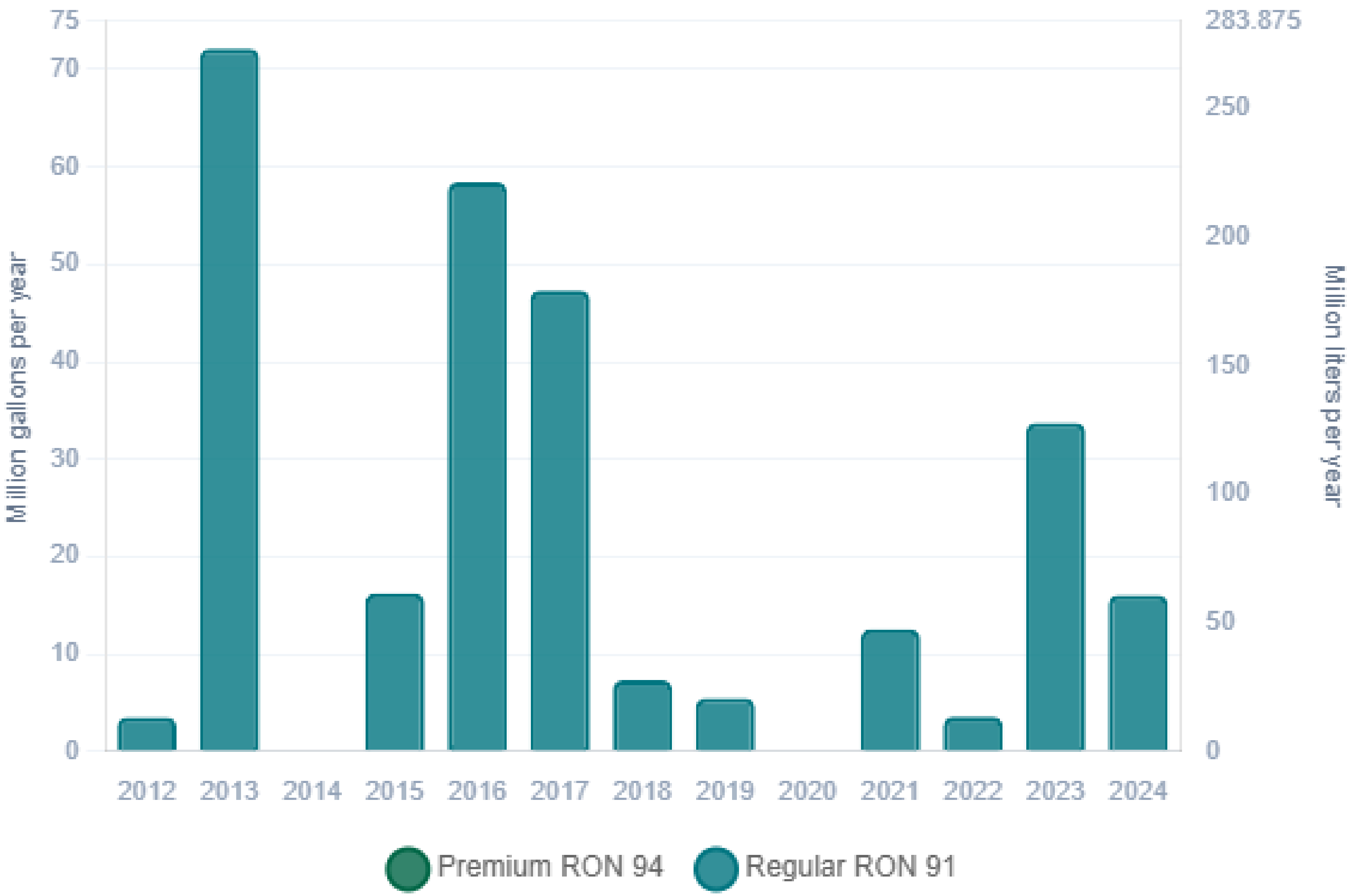
Source: Korean Energy Economics Institute

Gasoline Production in South Korea



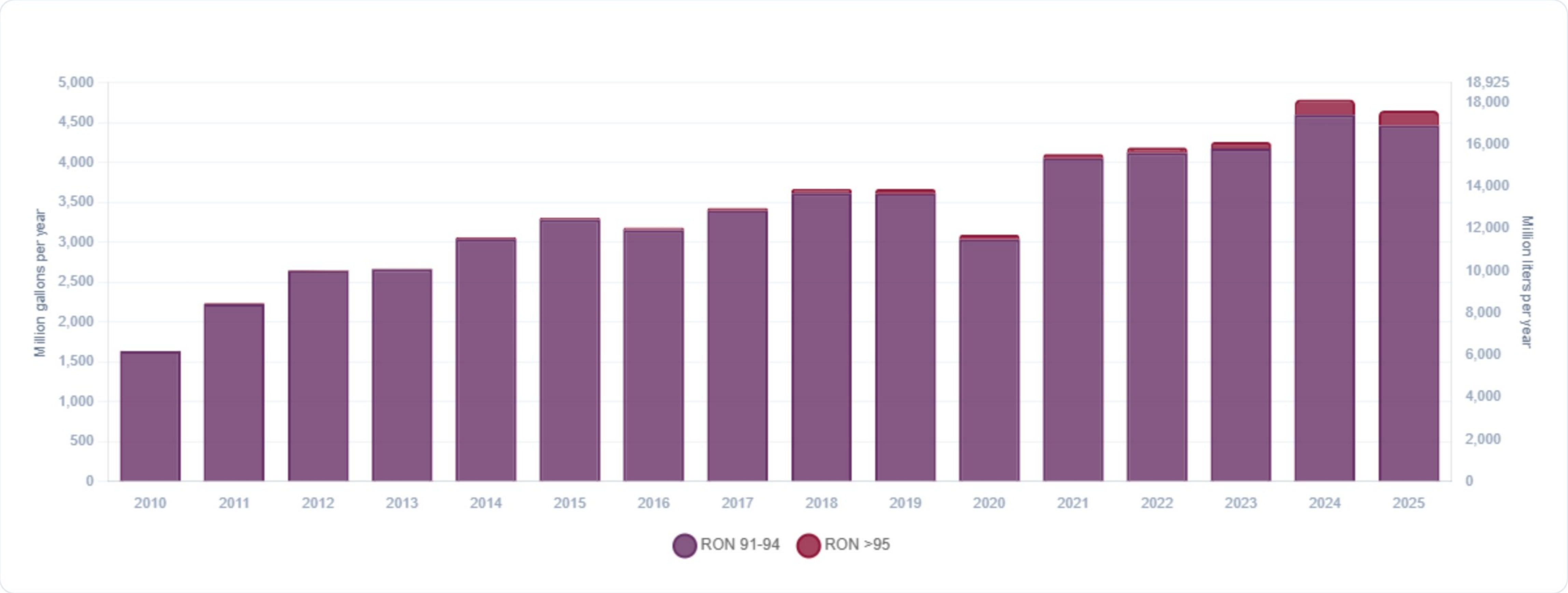
Source: Korean Energy Economics Institute

Gasoline Imports in South Korea



Source: Korean Energy Economics Institute

Gasoline Exports in South Korea



Source: Korean Energy Economics Institute

Ethanol Balance in South Korea

South Korea does not produce or consumes bioethanol in a commercial level for gasoline blending. It imports large quantities of bioethanol for industrial uses.

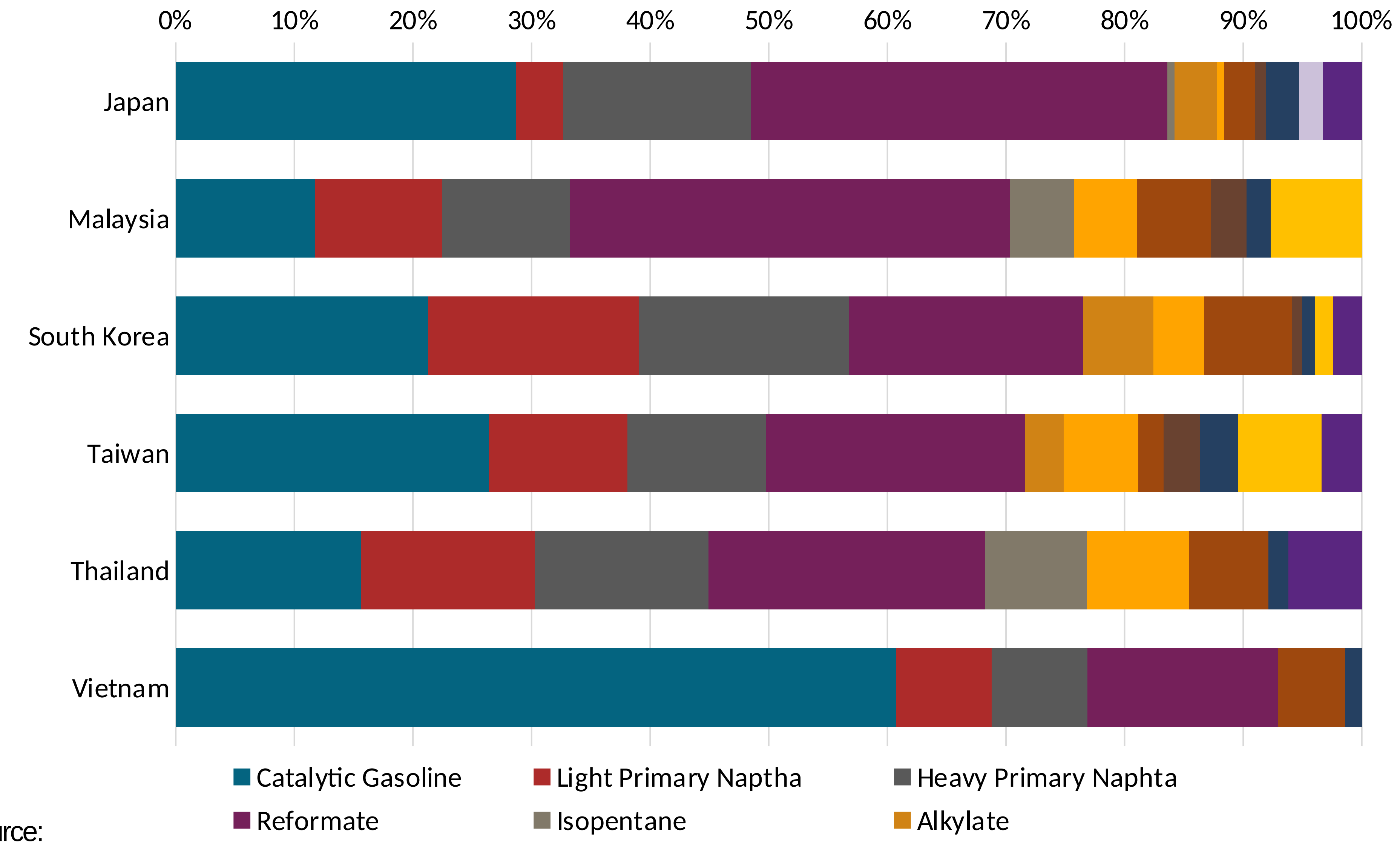
Gasoline Quality in South Korea

Name	Clean Air Conservation Act			
Implementation Year	2009			
Applicability	Nationwide			
Grade	RON 91		RON 94	
	Min	Max	Min	Max
RON	91.0		94.0	
MON				
AKI				
Benzene (%vol)		0.7		0.7
Aromatics (% vol)		24.0		24.0
Olefins (%vol)		18.0		18.0
Lead (g/L)		0.013		0.013
Manganese (mg/L)		-		-
Sulfur (ppm)		10		10
Oxygen (%v)	0.5	2.3	0.5	2.3
Ethanol (%vol)				
MTBE (%vol)				
RVP (psi)		8.7		8.7

Gasoline Component Blending in Asia

Gasoline is a blend of a base gasoline and other components. Each component provides different properties to the final blend, for example, isomerates, alkylates and butanes increase the octane. The components commonly used in Asia are:

- Catalytic Gasoline
- Light Primary Naptha
- Heavy Primary Naphta
- Reformate
- Isopentane
- Alkylate
- Isomerate
- Hydrocracked Gasoline
- Coker Naphta
- n-Butane
- MTBE
- ETBE
- TAME
- Aromatics



Source: Faro90

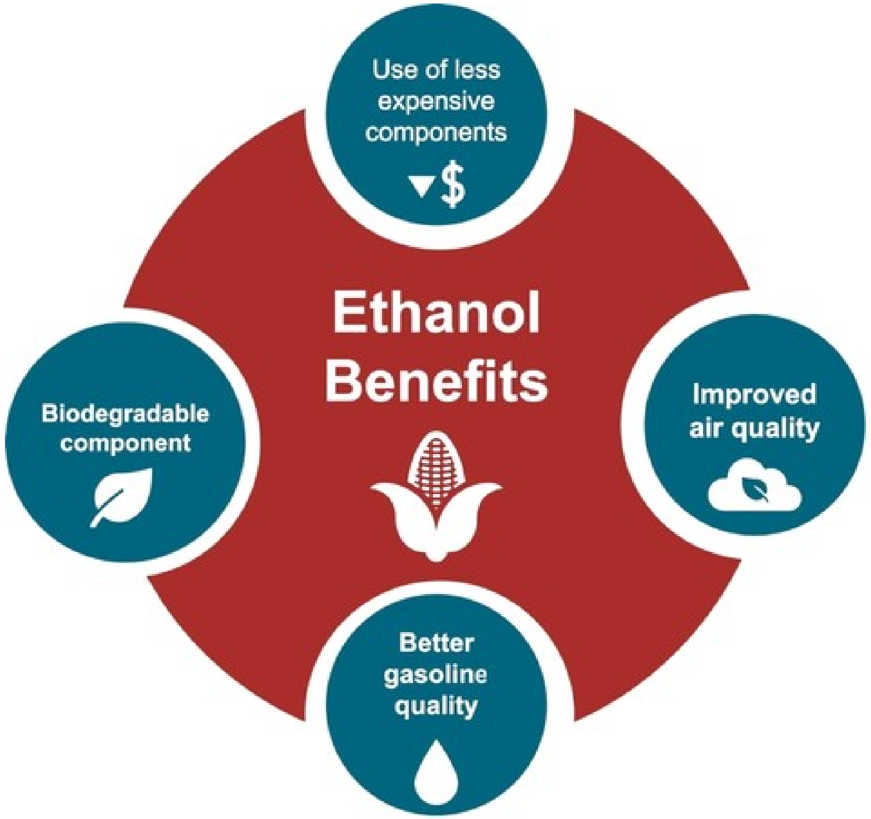
Gasoline Blending Optimization

In some parts of the world, ethanol is added to gasoline as a blending component.The advantages of ethanol include that it is a renewable fuel made of biomass; that it is an octane booster that helps to dilute sulfur; and that it allows the fulfillment of environmental objectives.To determine the optimal components to be blended with ethanol, a **blending model** was used. This model selects the components to add in the gasoline/ethanol blend based on:

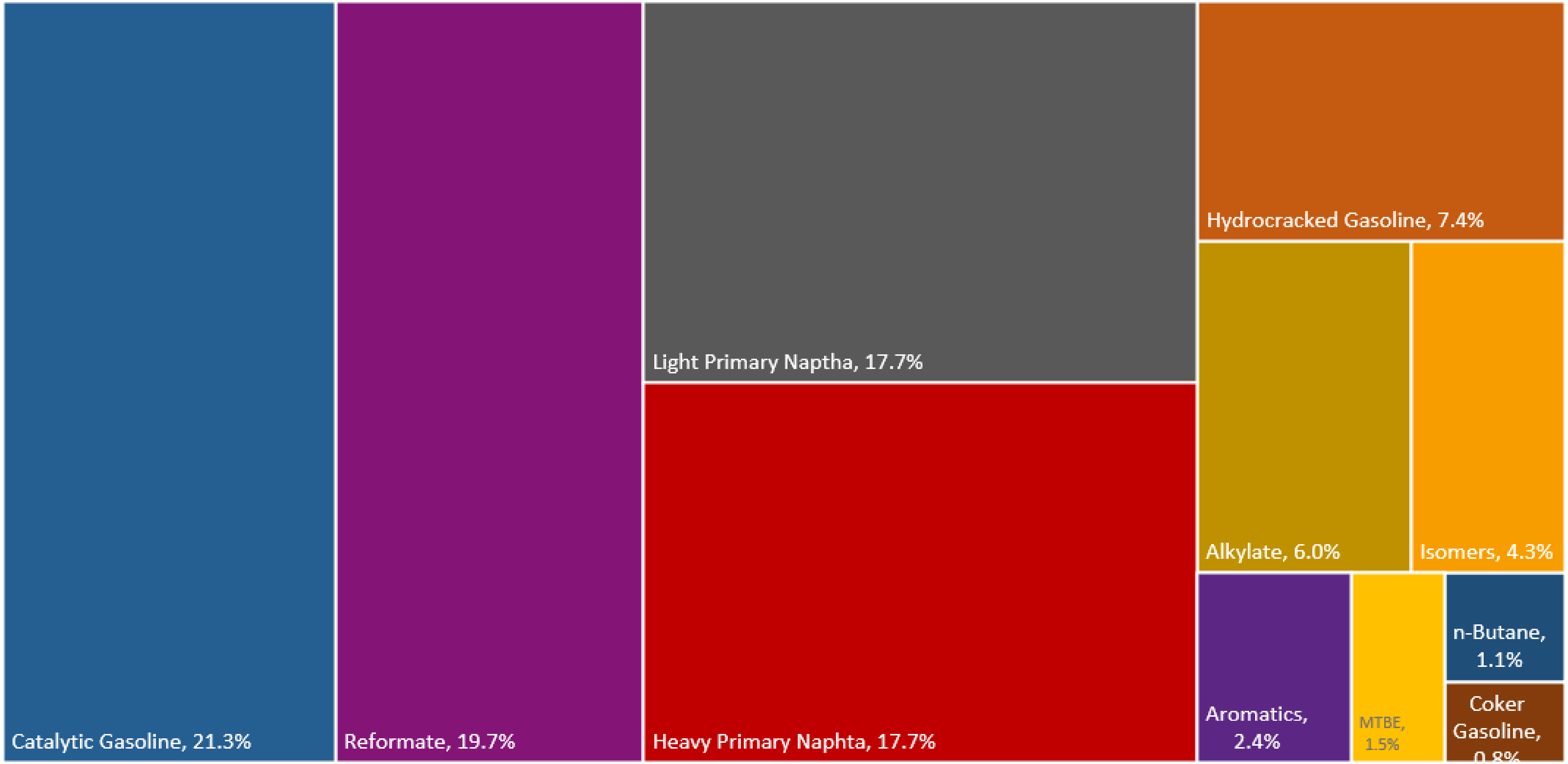
- Components prices,
- Properties each component affects,
- Quality parameters by country, and
- Component availability by country.

Through iterations, the model obtains the %v/v of the components to be blended with 10%, 15%, 20%, 25% and 30% of ethanol, in such a way that the final blend complies with the required properties of a finished gasoline by country.

The blending model uses gasoline component spot average prices 2025 and provides fuel prices that do not include country distribution costs, local taxes and subsidies and import or gas station margins.

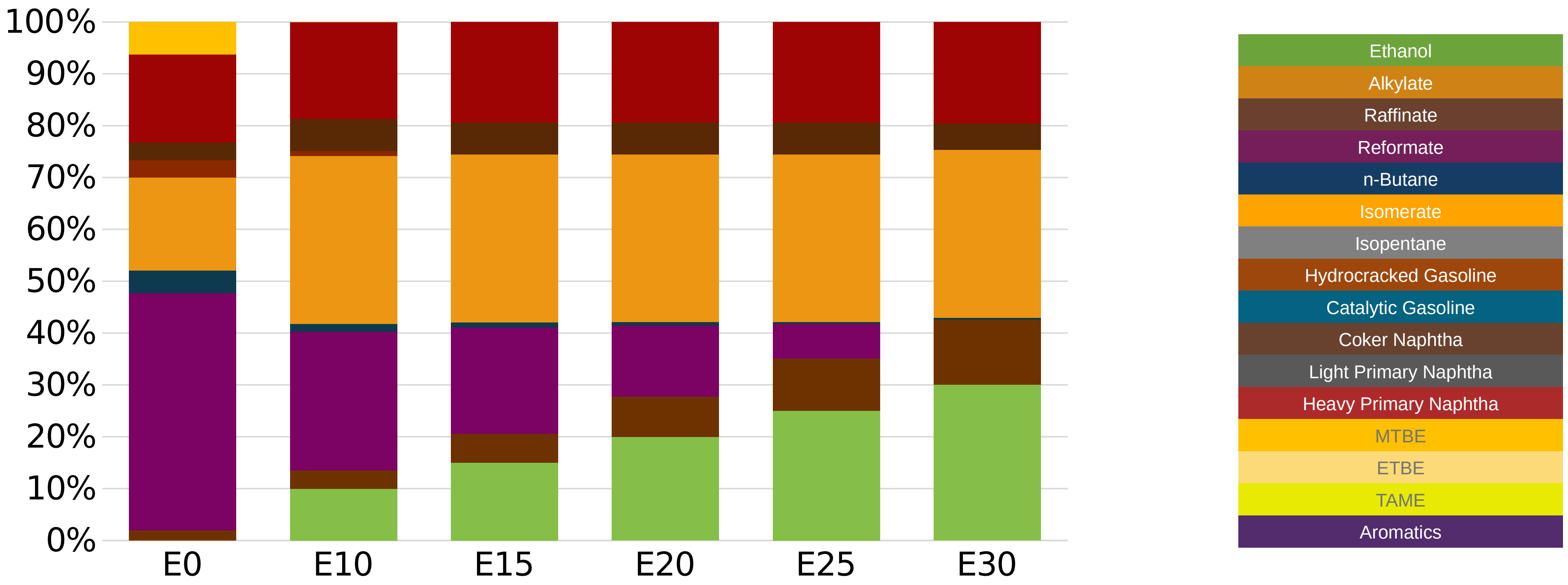


South Korea Available Blending Components



Source:
Faro90

South Korea – Gasoline 91 – Constant Octane

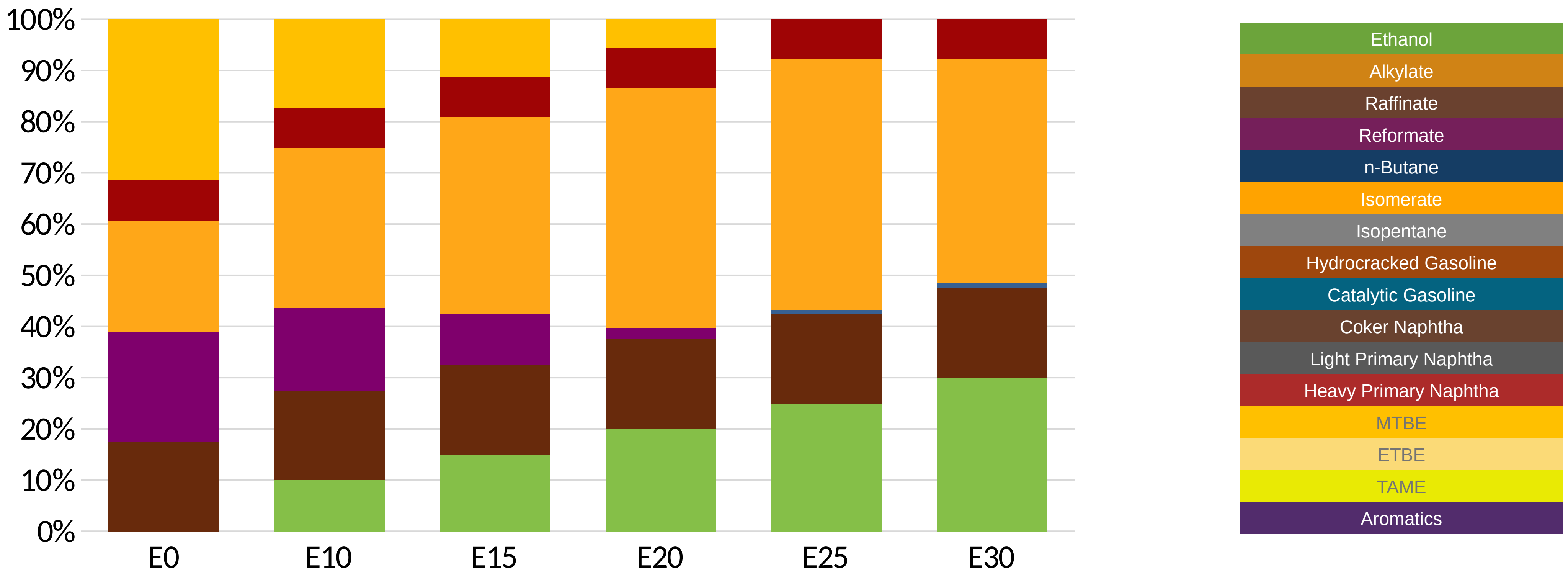


Octane (RON)	91.0	91.0	92.1	93.2	94.3	95.3
Price (US$/gal)	$2.03	$1.97	$1.93	$1.87	$1.85	$1.87

Average CIF 2025 Prices

Source: Faro90

South Korea - Gasoline 94 - Constant Octane

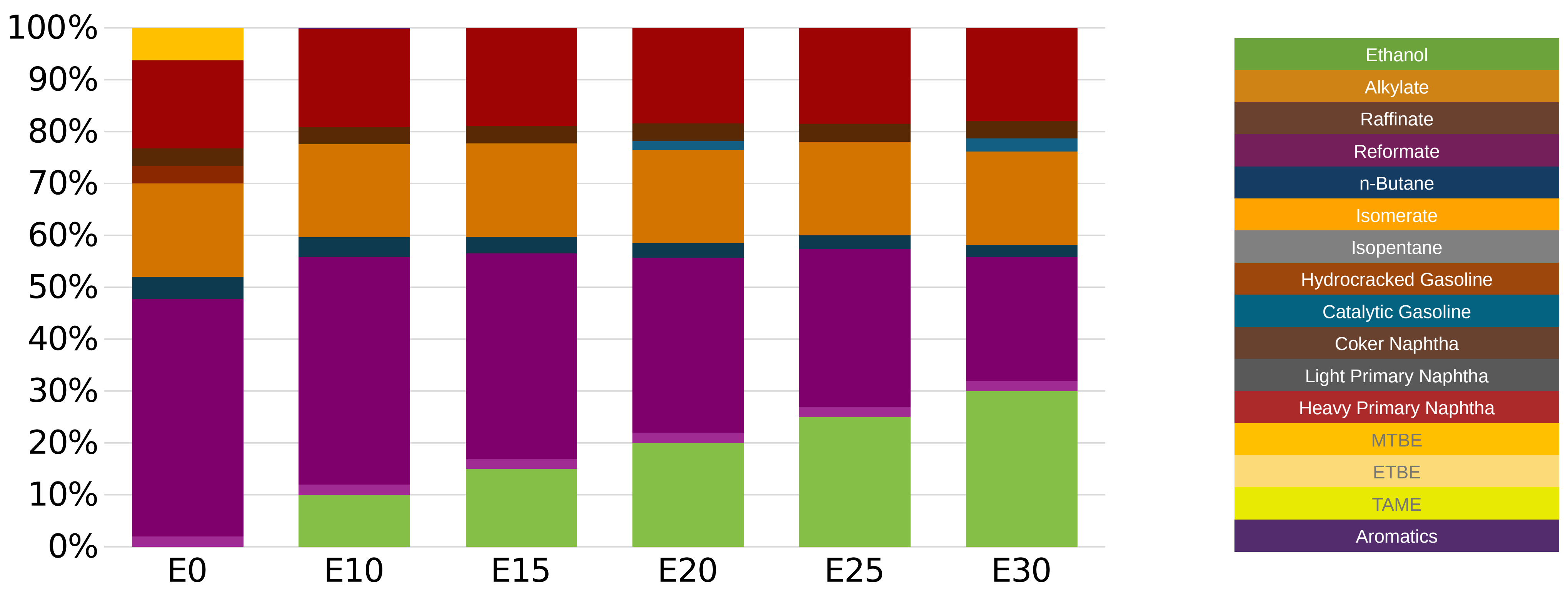


Octane (RON)	94.0	94.0	94.0	94.0	94.7	96.9
Price (US$/gal)	$2.04	$1.92	$1.88	$1.84	$1.79	$1.74

Average CIF 2025 Prices

Source: Faro90

South Korea – Gasoline 91 – Increased Octane

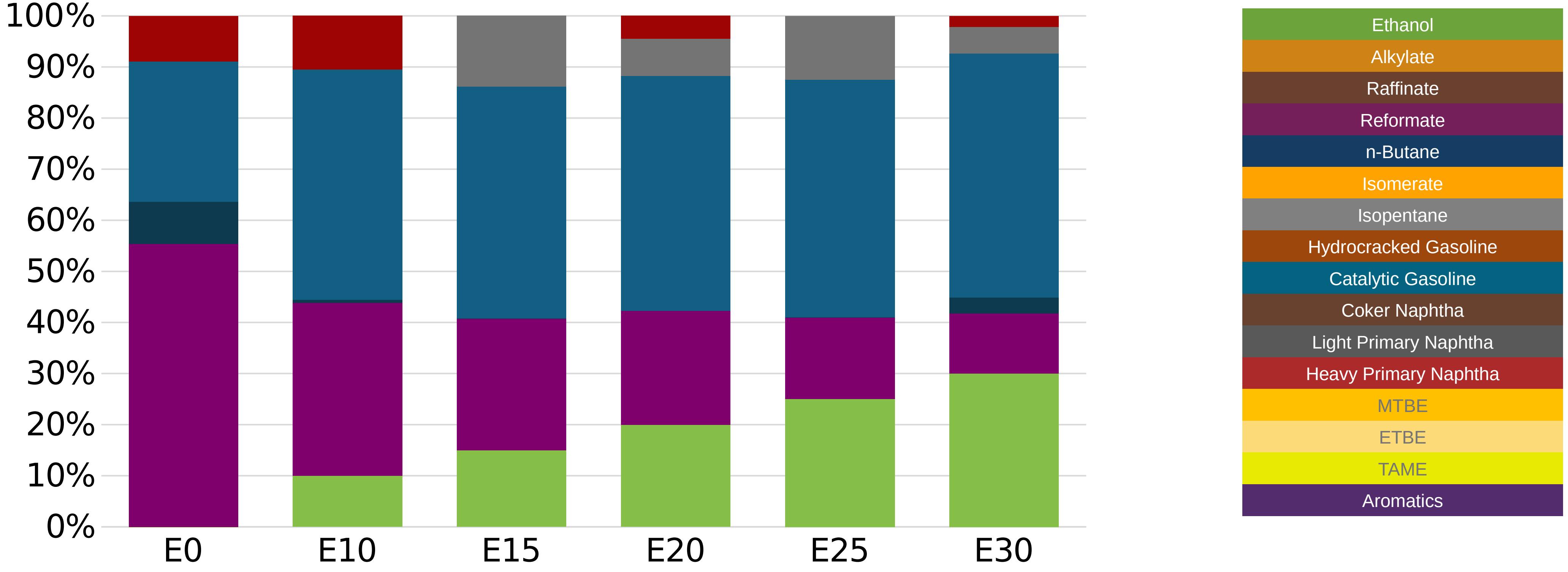


Octane (RON)	91.0	93.7	95.4	97.1	98.8	100.3
Price (US$/gal)	$2.03	$2.03	$2.03	$2.03	$2.03	$2.03

Average CIF 2025 Prices

Source: Faro90

South Korea – Gasoline 94 – Increased Octane



Octane (RON)	94.0	94.0	94.2	95.6	97.3	99.5
Price (US$/gal)	$2.04	$2.04	$2.04	$2.04	$2.04	$2.04

Average CIF 2025 Prices

Source: Faro90

Vehicle Emission Impact for Ethanol Gasoline Blending

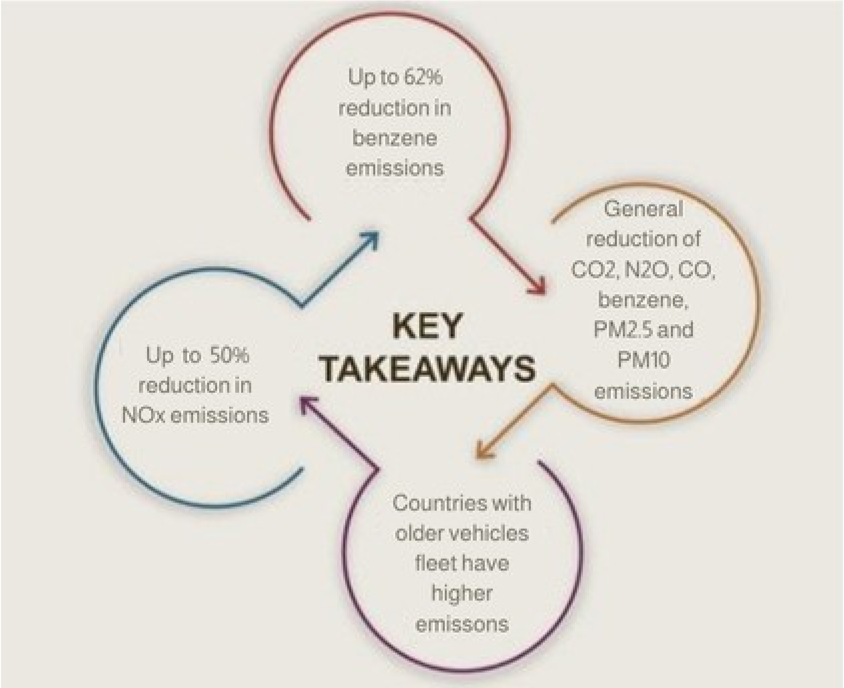
The model used in this analysis takes as a reference the **International Vehicle Emissions Model (IVE).**
The model uses the Base Emission Rates from IVE model, as well as its Adjustment Factors based on:

- Vehicle technology (cars, trucks, buses, motorcycles),
- Vehicle fleet average age,
- Average traveled distance per vehicle by country, as well as
- Geographical and climatic conditions (altitude, humidity, temperature).

Emissions of criteria pollutants, toxic pollutants, and greenhouse gases (GHG) were calculated and calibrated with emission inventories, using real gasoline quality data. The reduction rates for gasoline/ethanol blends were obtained from various sources (IPCC, US Grains, among others).

Emission estimations for different pollutants for gasoline and gasoline/ethanol blends (10%, 15%, 20%, 25% and 30% ethanol) were determined using the IVE Model. A comparison between the results and the European (Euro 6) requirements is made. Results are also compared with real emissions of the United States vehicle fleet*.

Main Results



**Source: Bureau of transportation statistics.*

Gasoline Vehicle Fleet - South Korea

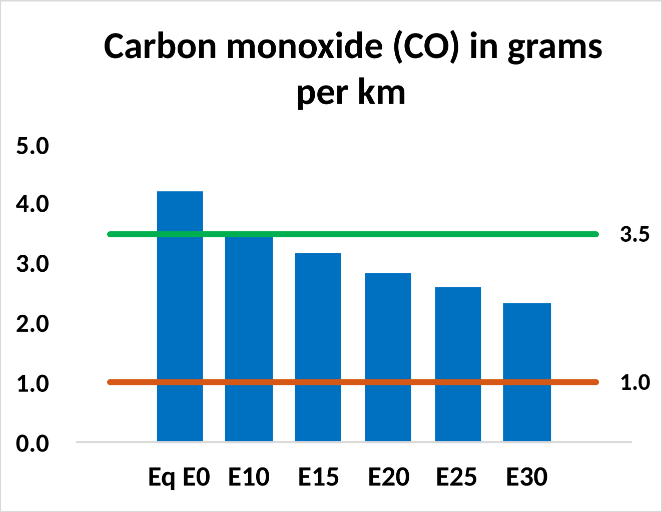
Type	Weight	Engine	Exhaust	Recovery	Age	No. Vehicles
Auto/Sml Truck	Light	Carburator	3-Ways	PCV Tank	>8 years	60,543
		Single FI	3-Ways		>8 years	302,966
		Multi FI	3-Ways		>8 years	887,895
			Euro III		>8 years	999,787
			Euro IV		>8 years	1,720,399
			Euro V		>8 years	590,827
			Euro 6		>8 years	1,870,698
					4-8 years	1,628,723
					<4 years	1,325,367
			Hybrid		All	635,201
		Electric			All	226,384
	Medium	Carburator	3-Ways	PCV Tank	>8 years	90,814
		Single FI	3-Ways		>8 years	454,449
		Multi FI	3-Ways		>8 years	1,331,843
			Euro III		>8 years	1,499,681
			Euro IV		>8 years	2,580,598
			Euro V		>8 years	886,240
			Euro 6		>8 years	2,806,047
					4-8 years	2,443,085
					<4 years	1,988,051
			Hybrid		All	952,801
		Electric			All	339,576
Sml Engines	Light	4-Cycle	None	PCV	>8 years	131,266
			Euro III		>8 years	191,007
			Euro IV		>8 years	487,200
			Euro V		>8 years	153,563
			Euro 6		>8 years	476,343
					4-8 years	422,096
					<4 years	496,675
			Hybrid		All	159,568
		Electric			All	56,870
Truck	Heavy	Carburator	3-Ways	PCV	>8 years	43,249
		Fuel-Injection	3-Ways		>8 years	506,447
			Euro III		>8 years	130,446
			Euro IV		>8 years	326,972
			Euro V		>8 years	157,414
			Euro 6		>8 years	933,427
					4-8 years	1,040,629
					<4 years	944,962
			Hybrid		All	276,910
		Electric			All	98,690

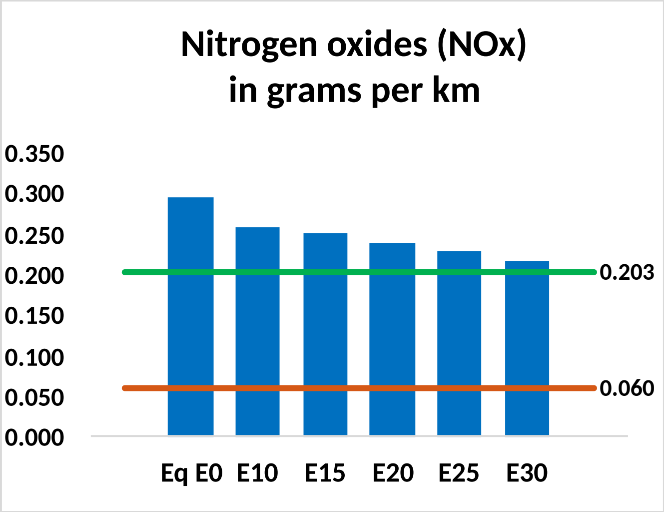
Vehicle Fleet: 32,655,710
Gasoline Vehicle Fleet: 27,483,451

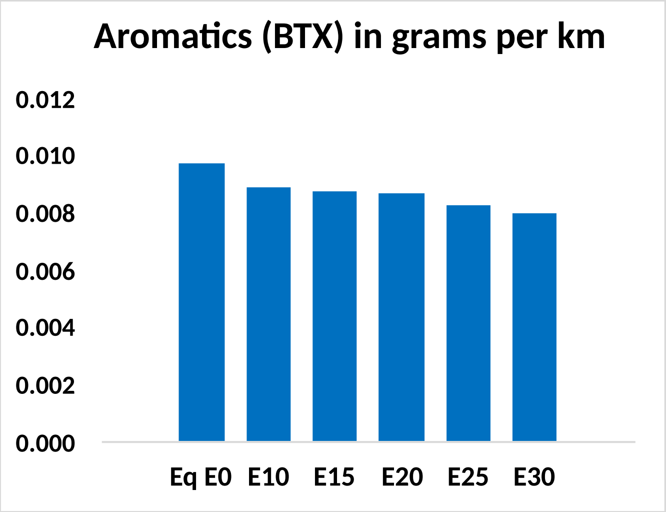
Source: MOLIT

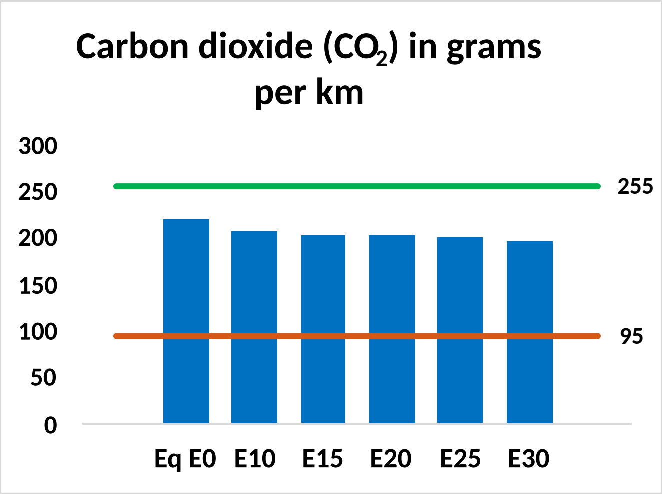
South Korea – Vehicular Emissions

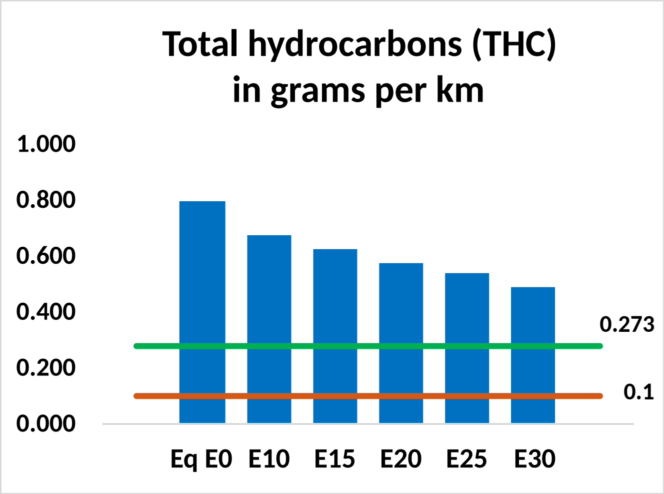
TIER III BIN 125 United States
Euro 6 Standard

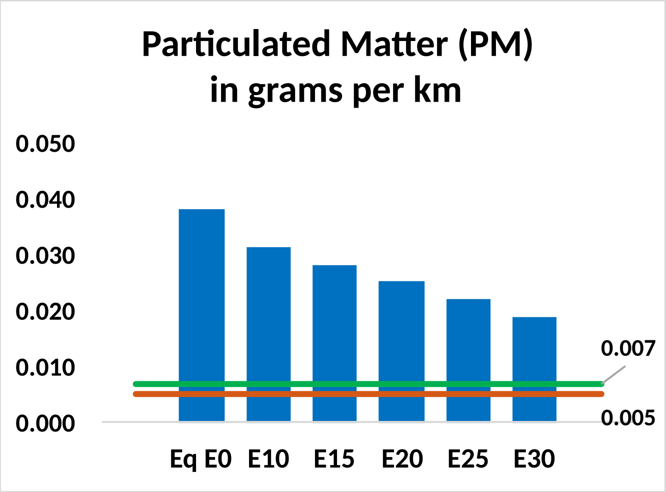












Source: Faro90

South Korea – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	2,698,816	2,459,685	2,220,554	2,018,419	1,812,161	1,652,704	1,484,377	-18%	-33%
CO2	139,947,032	136,348,394	132,949,680	130,276,692	128,953,993	128,324,905	125,959,508	-5%	-8%
VOC	507,796	469,392	430,988	399,641	368,152	343,957	311,608	-15%	-27%
NOx	189,582	175,047	165,224	160,298	152,275	146,026	138,644	-13%	-20%
SOx	1,639	1,515	1,391	1,286	1,186	1,077	962	-15%	-28%
NH3	50,231	49,734	49,236	49,470	50,027	50,418	50,743	-2%	0%
N2O	2,899	2,883	2,872	2,885	2,945	2,978	3,012	-1%	2%
CH4	105,041	105,041	105,041	106,617	108,928	110,924	112,814	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	2,100	1,986	1,873	1,789	1,710	1,649	1,565	-11%	-19%
Acetaldehyde	6,839	7,849	11,517	17,539	23,895	27,304	32,263	68%	249%
Formaldehyde	26,080	25,355	29,557	34,706	36,360	40,224	43,788	13%	39%
Benzene	6,223	5,980	5,667	5,577	5,531	5,289	5,119	-9%	-11%
PM2.5	7,880	7,164	6,447	5,817	5,200	4,539	3,847	-18%	-34%
PM10	16,575	15,069	13,562	12,236	10,938	9,547	8,092	-18%	-34%

Source: Faro90

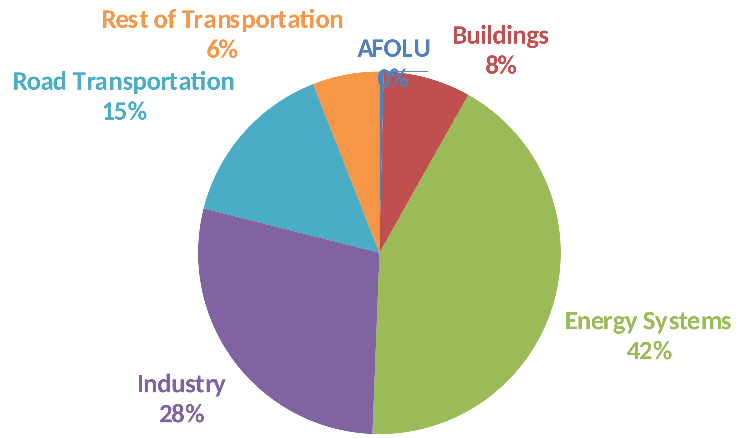
Life Cycle GHG Emissions

The Global Agenda for Climate Action calls for countries, cities and regions, businesses, and members of civil society around the world to to achieve a low-carbon and climate-resilient economies in support of the Paris Agreement. Road Transportation accounts for 15% of total GHG emissions (IPCC,2023).

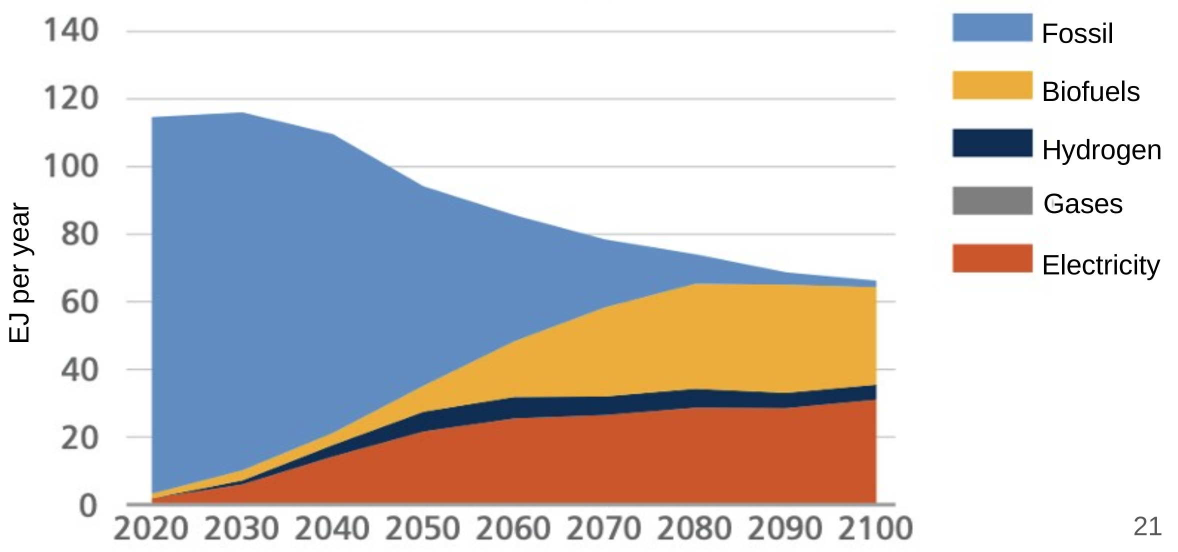
Ethanol gasoline blends are an internationally adopted practice that offer significant short-term benefits. Biofuels are an important solution to achieve reduction targets in the transportation sector.

Country GHG emission reductions are calculated using **RED II/III** and **GREET** life cycle emission methodologies and the **International Vehicular Emissions Model (IVE).** Results are compared to the current country targets for 2035. Current 2025 country emission are those published by **IPCC**.

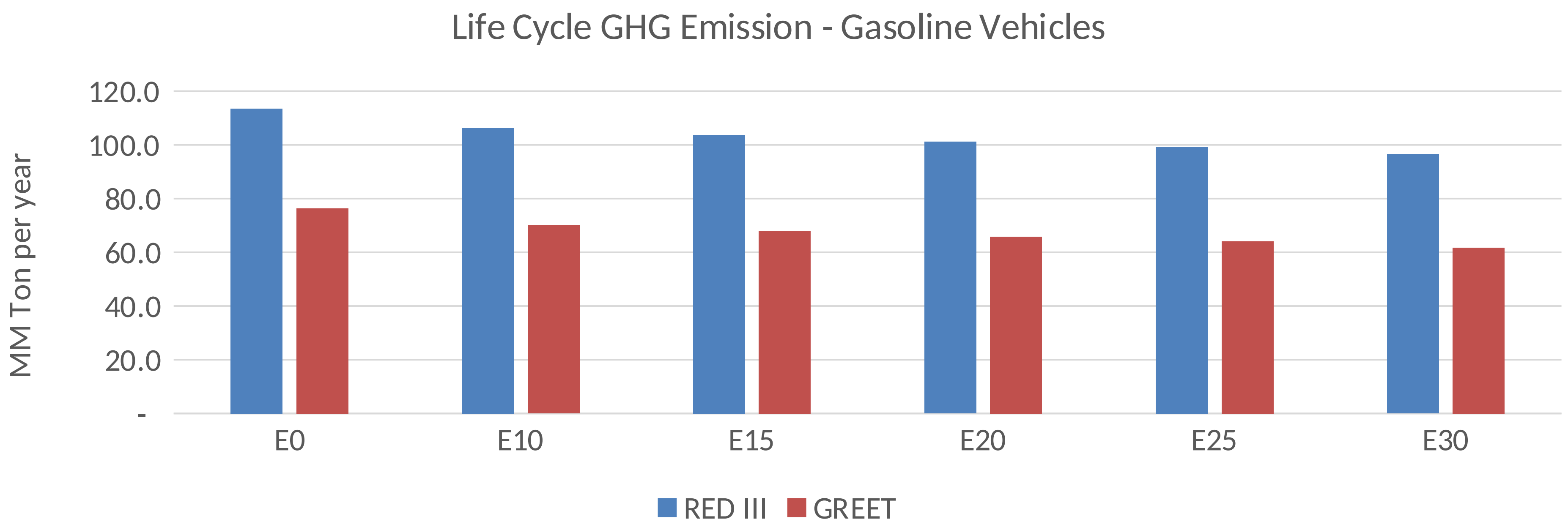
Current GHG Global Emissions Distribution



IPCC Transport Sector Sustainable Development Goals and climate policies Scenario, 2023



South Korea – Gasoline Vehicle Fleet Life Cycle GHG Emissions



Country Emissions 2024 (MMT/año)	682.3
Target @ 2035 (MMT/year)	332.3
Delta	349.9

Source: Greet, RED II/III, climateactiontracker.org, F90